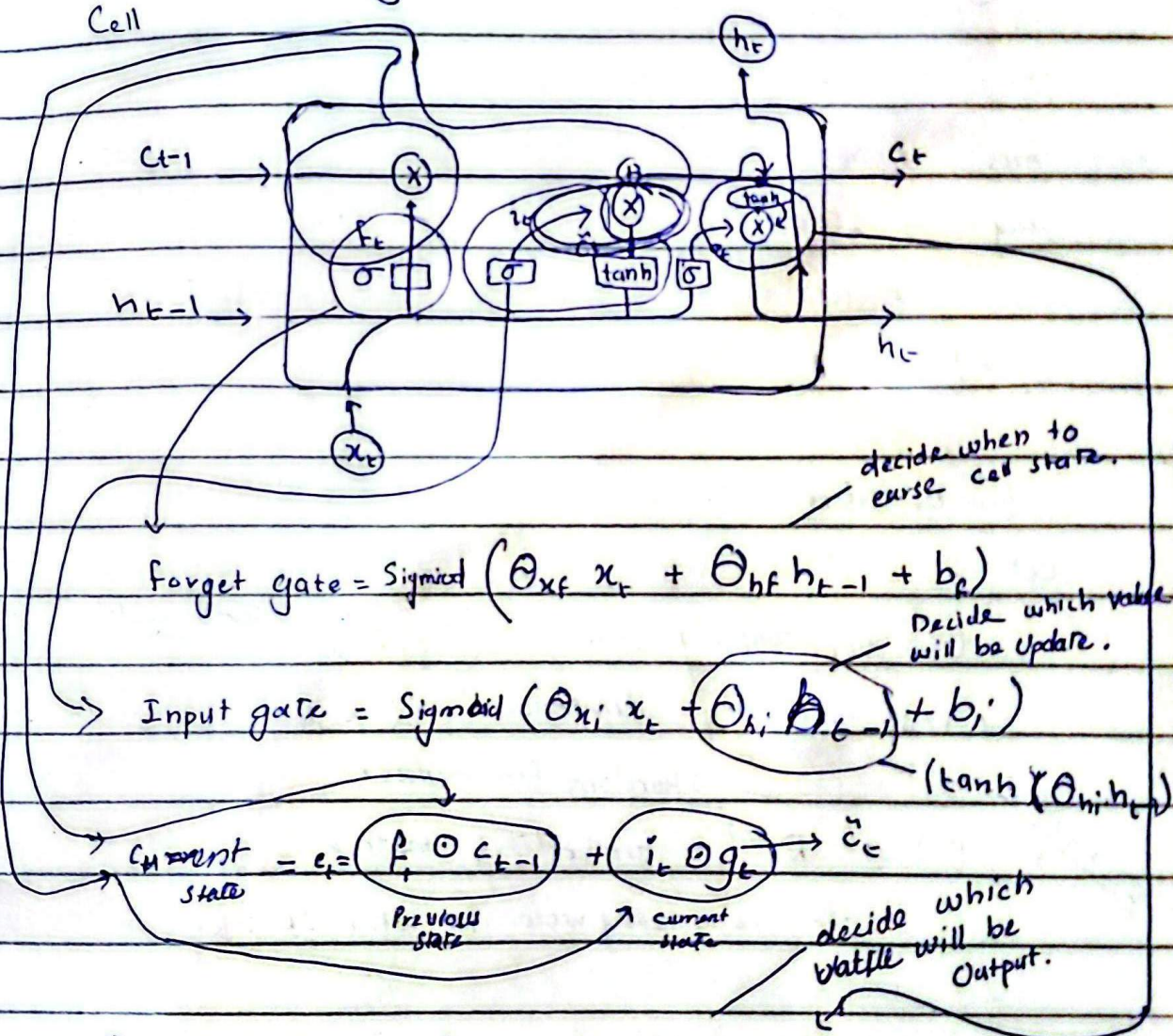


$\Rightarrow$ Long term Vanishing gradient

$$\tanh(\text{ANN}) = \begin{bmatrix} 0.379949 \\ -0.0798298 \end{bmatrix} = x_{t+1}$$

"Using LSTM"



Output gate = $h_t = o_t \odot \tanh(c_t)$

Candidate cell = $\tilde{c}_t$

Cell Update = $g_t = \tanh(\theta_{xg} x_t + \theta_{hg} h_{t-1} + b_g)$

$\tanh(-1, 1)$

Transformers : Attention and Self attention

Embedding

tokens :

[CLS] $\rightarrow$ 101 -

today $\rightarrow$ 2651 -

is $\rightarrow$ 2003 -

a $\rightarrow$ 1037

nice $\rightarrow$ 3835

day $\rightarrow$ 2154

. $\rightarrow$ 1012

[SEP] - 102

Embedding layer

Let =

(V) Vocabulary size = (30522) ^{BERT-base}

(D) embedding dimensions $\rightarrow$ (768)

$\leftarrow (E) \in \mathbb{R}^{V \times D}$ = embedding matrix.
 ²³⁴⁴⁰⁸⁹⁶ $\rightarrow$ Random Real numbers.

Row i = embedding vector for token ID i

Formula :

token ID is t_i

$$v_i = E[t_i]$$

$v_i \in \mathbb{R}^D$ = vector representation of token t_i

Essentially: Look up row t_i in the embedding

$\rightarrow$ where positional embedding. ^{Matrix}

$$P \in \mathbb{R}^{L \times D}$$

where L = sequence length
 D = Dimension of vectors

Positional Embedding vector = $E[\text{tokens}] + P[\text{index of token}]$

create $V \times D$ of the positional embedding

vector of each token

Suppose: Positional Embedding Layer vectors (h)

$$H = [h_1, h_2, \dots, h_L]$$

Goal, Let look at "all other tokens in the Sequence and decide how much attention to pay to each when updating its representation

Result: Context-aware vectors for each token.

I'm token (3) (nice). want to update vector based on how important every other token (Today, is, a, day) to me.

Formula Self-Attention:

for each token embedding $h_i \in \mathbb{R}^D$

$$q_i = W_Q h_i, \quad k_i = W_K h_i, \quad v_i = W_V h_i$$

where

- $W_Q, W_K, W_V \in \mathbb{R}^{D \times d_h} \rightarrow$ learn metrics.
- d_h is the dimension of the query/key vectors (can be same as D or smaller).
- q_i = query vector for token i
- k_i = Key vector for token i
- v_i = Value vector for token i .

→ Computation Attention Scores: (Compatibility) ✓

$$\text{Attention} = \text{Softmax} \left(\text{Score} \left(\frac{q_i \cdot k_j}{\sqrt{d_k}} \right) \right) \cdot V_j$$

→ Dot product b/w q_i and k_j measures how much token i should pay attention to token j .

→ Divide by $\sqrt{d_k}$ → Scale Factor to avoid larger value

→ Applying Softmax to get attention weight.

→ The U_i^p vector contain new contextual info about from the entire sequence.

position		
Example	Tokens	Embedding h_i
1	→ today	1, 0, 1, 0
2	→ is	0, 1, 0, 1
3	→	1, 1, 0, 0
4	nice	0, 1, 1, 0
5	day	1, 0, 0, 1
6	[SEP]	0, 0, 1, 1

Embedded dimension:

$D = 4$, seq - L

$$h = \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 1 & 1 & 0 & 0 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 0 & 1 & 1 \end{bmatrix}$$

Q, K, V ?

$$W_Q, W_K, W_V = \in \mathbb{R}^{h \times h} \text{ (4x4)}.$$

So,

$$Q = HW_Q = H$$

$$K = HW_K = H$$

$$V = HW_V = H$$

Let Compute Attention for Nice

where

$$q_{\text{nice}} = [0, 1, 1, 0]$$

$$d_k = 4 = \sqrt{4} = 2$$

$$\text{score} = \left(\frac{Q K^T}{\sqrt{d}} \right) \quad \text{only for nice.} \quad - \quad \underbrace{(1 \times 4) (4 \times 1)} = (1 \times 1)$$

Token =

Dot product

today

$$1 \times 0 + 0 \times 1 + 1 \times 1 + 0 \times 0 = 1/2 = 0.5$$

is

$$0 \times 0 + 1 \times 1 + 0 \times 1 + 1 \times 0 = 1/2 = 0.5$$

q

$$1 \times 0 + 1 \times 1 + 0 \times 1 + 0 \times 0 = 1/2 = 0.5$$

nice

$$0 \times 0 + 1 \times 1 + 1 \times 1 + 0 \times 0 = 2/2 = 1$$

day

$$0 \times 0 + 0 \times 1 + 0 \times 0 + 1 \times 0 = 0$$

[~~stop~~]

$$0 \times 0 + 0 \times 1 + 1 \times 1 + 1 \times 0 = 1/2 = 0.5$$

Applying Softmax:

$$\frac{e^{s_i}}{\sum e^{s_i}}$$

Probability $\left[0.15, 0.15, 0.15, 0.25, 0.08, 0.15 \right] = \text{sum} = 1$

Attention = Attention score $\times V_j$

$$= 0.15 \times (1, 0, 1, 0) + 0.15 (0, 1, 0, 1) \\ + 0.15 \times (1, 1, 0, 0) + 0.25 (0, 1, 1, 0) \\ + 0.25 \times (1, 0, 0, 0) + 0.15 \times (0, 0, 1, 1)$$

today

today	[0.15, 0.15, 0.15, 0.15]
is	[0, 0.15, 0, 0.15]
a	[0, 0.15, 0.15, 0]
nice	[0, 0.25, 0.25, 0]
day	[0.08, 0, 0, 0.08]
[SEP]	[0, 0, 0.15, 0.15]

add
dimension
vice.

$$\text{nice} = (0.15 + 0.15 + 0.08, 0.15 + 0.15 + 0.25, 0.15 + 0.25 + 0.15, \\ 0.15 + 0.08 + 0.15)$$

$$\text{nice} = (0.38, 0.55, 0.55, 0.33)$$

$$\text{nice} = (0.38, 0.55, 0.55, 0.33)$$

Now Nice token added to all words.

Positional Encoding formula: from (Transformer)

$$PE(pos, 2i) = \sin \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right)$$

$$PE(pos, 2i+1) = \cos \left(\frac{pos}{10000^{\frac{2i}{d_{model}}}} \right)$$

$pos \rightarrow$ Position of the token in the sentence

$i \rightarrow$ Dimension Index.

$2i/2i+1 \rightarrow$ even/odd embedding dimension

$d_{model} \rightarrow$ Embedding size. $\rightarrow 768$

$10000 \rightarrow$ Scaling constant for frequency control.

Input embedding $\rightarrow$ Positional Embedding $\rightarrow$ token Embedding

$\Rightarrow$ from previous $\rightarrow$ table $\rightarrow$ nice position is 4 but we will consider it 3 because I forgot the 0 in table.

Since $D = 4$

Dimension index

$$0 \quad \sin(\text{pos}/10000^0) = \sin(\text{pos})$$

$$1 \quad \cos(\text{pos}/10000^0) = \cos(\text{pos})$$

$$2 \quad \sin(\text{pos}/10000^{1/2}) = \sin(\text{pos}/100)$$

$$3 \quad \cos(\text{pos}/10000^{1/2}) = \cos(\text{pos}/100)$$

~~4~~

for Nice, $\text{pos} = 3 \rightarrow \text{token}(\text{nice})$:

Not degree

$$\sin(3) \xrightarrow{\text{Radians}} = 0.141$$

$$\cos(3) = -0.98995$$

$$\sin(3/100) = 0.029955$$

$$\cos(3/100) = 0.999555$$

$$\text{Seq. length} = 5$$

$$\text{Embedding dimension} = 4$$

$$\text{Number of head} = 2$$

$$\text{Head dimension} = d-k = 4/2 = 2$$

New Numerical

Seq Length = 3

Embeddy = 4

head

head-d = $\frac{2}{4/2} = 2$

Suppose

Pos token

Embeddy (40)

0 today 1 0 1 0

1 is 0 2 0 2

2 nice 1 1 1 1

$$X = \begin{bmatrix} & 0 & 1 & 2 & 3 \\ 1 & 0 & 1 & 0 \\ 0 & 2 & 0 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix}$$

Position Encoding

$$i=0 \quad \frac{2^i}{4} = \frac{10000}{10000} = 1$$

$$i=1 \quad \frac{2^i}{4} = \frac{10000}{10000} = 1$$

for today

$$\begin{aligned} 0 \quad \sin(\text{Pos}) &= \sin\left(\frac{\text{Pos}}{10000}\right) = \sin\left(\frac{0}{10000}\right) = 0 \\ 1 \quad \cos(\text{Pos}) &= \cos\left(\frac{\text{Pos}}{10000}\right) = \cos\left(\frac{0}{10000}\right) = 1 \\ 2 \quad \sin(\text{Pos}/100) &= \sin\left(\frac{0}{100}\right) = 0 \\ 3 \quad \cos(\text{Pos}/100) &= \cos\left(\frac{0}{100}\right) = 1 \end{aligned}$$

for nice

$$\begin{aligned} 0 & \quad \sin\left(\frac{1}{10000}\right) = 0.00009999833443267 \\ 1 & \quad \cos\left(\frac{1}{10000}\right) = 0.99999999833443267 \\ 2 & \quad \sin\left(\frac{1}{100}\right) = 0.009999833443267 \\ 3 & \quad \cos\left(\frac{1}{100}\right) = 0.9999833443267 \end{aligned}$$

for is

$$\begin{aligned} 0 & \quad \sin\left(\frac{2}{10000}\right) = 0.00019999666886534 \\ 1 & \quad \cos\left(\frac{2}{10000}\right) = 0.99999999666886534 \\ 2 & \quad \sin\left(\frac{2}{100}\right) = 0.01999833443267 \\ 3 & \quad \cos\left(\frac{2}{100}\right) = 0.999833443267 \end{aligned}$$

$$x_{\text{final}} = X + PE$$

for today

$$\begin{bmatrix} 1 & 0 & 1 & 0 \end{bmatrix} + \begin{bmatrix} 0.1 & 0 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \end{bmatrix}$$

for is

$$\begin{bmatrix} 0 & 2 & 0 & 2 \end{bmatrix} + \begin{bmatrix} 0.84 & 0.54 & 0.01 & 0.995 \end{bmatrix}$$

for Nice:

$$\begin{bmatrix} 1 & 1 & 1 & 1 \end{bmatrix} + \begin{bmatrix} 0.90930 & -0.41615 & 0.0200 & 0.99980 \end{bmatrix}$$

$$X_F = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0.841 & 2.540 & 0.0100 & 2.9955 \\ 1.90930 & 0.58325 & 1.0200 & 1.99980 \end{bmatrix}$$

Now 2 head

$$X_1 = \begin{bmatrix} 1 & 1 \\ 0.841 & 2.540 \\ 1.909 & 0.583 \end{bmatrix} \quad X_2 = \begin{bmatrix} 1 & 1 \\ 0.010 & 2.995 \\ 1.020 & 1.99980 \end{bmatrix}$$

$$QX_i = V X_i = K X_i = I$$

Scaled Dot product Attention

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right) \cdot V$$

here, $\sqrt{d_k} = \sqrt{2} = 1.414$.

$$Q = (3 \times 2) \rightarrow K = (2 \times 3)^T = (3 \times 3)$$

Suppose

$$QK^T = \begin{bmatrix} 2 & 3.38 & 2.49 \\ 3.38 & 7.15 & 3.07 \\ 2.49 & 3.07 & 3.1 \end{bmatrix}$$

$$\frac{QK^T}{1.414} = \begin{bmatrix} 1.414 & 2.391 & 1.763 \\ 2.391 & 5.062 & 2.178 \\ 1.763 & 2.178 & 2.018 \end{bmatrix}$$

~~Attention~~ softmax = rowwise.

$$(1) \quad \text{softmax}[1.414, 2.391, 1.763] = [0.193, 0.513, 0.281]$$

$$(2) \quad = [0.059, 0.859, 0.082]$$

$$(3) \quad = [0.175, 0.264, 0.561]$$

multiple with V

Attention

$$= \left(\frac{QK}{\sqrt{d_k}}\right) \cdot V = \begin{bmatrix} 1.344 & 1.751 \\ 0.940 & 2.311 \\ 1.410 & 1.296 \end{bmatrix}$$

~~head~~

Suppose same for head 2

$$K = (3 \times 2)$$

$$Q = (3 \times 2)$$

Mask will add in it

$$QK^T = (3 \times 2)(2 \times 3)$$

$$1.414 = (3 \times 3)$$

After Apply softmax

0.140	0.404	0.398	Mask
0.012	0.842	0.146	
0.691	0.091	0.145	

$$= \begin{bmatrix} 0.140 & 0.404 & 0.398 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

Softmax (QK^T)

Attention =

head 2

0.606	2.203
0.161	2.815
0.230	2.597

Concatenate = $(h_1; h_2)$

MHA(X) =

Mult head Attention

0.344	1.751	0.606	2.203
0.945	2.815	0.161	2.815
0.230	2.597	0.230	2.597

$$X = \begin{bmatrix} 1 & 1.71 & 1 \\ 0.8 & 2.5 & 0.06 & 2.995 \\ 1.909 & 0.583 & 1.020 & 1.7180 \end{bmatrix}$$

Residual

$$Z = X + MHA(X)$$

$$Z = \begin{bmatrix} 2.344 & 2.751 & 1.606 & 3.20 \\ .1.706 & 4.851 & 0.171 & 5.815 \\ 3.327 & 1.880 & 1.250 & 4.597 \end{bmatrix}$$

Apply Normalization
(~~per row~~) (per row)

$$\mu_1 = \text{mean}(z_{1,}) \text{ reduced}$$

$$\mu_2 = \text{mean}(z_{2,}) \text{ row}$$

$$\mu_3 = \text{mean}(z_{3,}) \text{ row}$$

$$\sigma = \sqrt{\frac{1}{d} \sum (z - \mu)^2}$$

$$N(z) = \frac{z - \mu}{\sigma}$$

row 1 (for today) token.

$$Z = [2.344 \quad 2.751 \quad 1.606 \quad 3.20]$$

$$\mu = 2.476$$

$$\sigma = 0.579$$

$$N(Z) = [-0.200, 0.475, -1.504, 1.229]$$

row 2 (for is) token

row 3 (for nice) token

~~token~~ +

output Masking + Residual + Normalization

$$H = \begin{bmatrix} -0.228 & 0.475 & -1.504 & 1.287 \\ -0.602 & 0.751 & -1.326 & 1.182 \\ 0.430 & -0.674 & -1.154 & 1.390 \end{bmatrix}$$

Linear

$$FFN(x) = \left(\max(0, xW_1 + b_1) \right) W_2 + b_2$$

$$W_1 = (4 \rightarrow 8) = (4 \times 8)$$

$$W_2 = (8 \rightarrow 4) = (8 \times 4)$$

(For today) tokens

$$Rel4 + xW_1 = [-1.730, 0.247, \dots, -0.029]$$

Applying ~~Rel4~~ ~~max~~ ~~Rel4~~ $\rightarrow$

$$= \begin{bmatrix} 0, 0.247, 1.029, 0, 0, 0.247 \\ 1.029, 0 \end{bmatrix}$$

for is token

$$[0, 0, 1.0, 0.22, 0, 0, 1.0, 0]$$

for is token = $[0, 0.144, 1.326, 0, 0, 0.144, 1.326, 0]$

~~is token~~

$$FFN(x) = [tokens_{all} W_2 + b_2]$$

$$y = \begin{bmatrix} 0, 0.494, 2.050, 0 \\ 0, 0.288, 2.652, 0 \\ 0, 0, 3.656, 0.440 \end{bmatrix}$$

Residual + Norm Layer

$$z = H + \text{FFN}(H) = \begin{bmatrix} -0.228, 0.969, 0.554, 1.252 \\ -0.607, 0.039, 1.326, 1.182 \\ 0.430, -0.64, 2.562, 1.848 \end{bmatrix}$$

$$N(z) = \left(\frac{z - \mu}{\sigma} \right)$$

$$\mu_1 = \text{mean}(\text{Row 1})$$

$$\mu_2 = \text{mean}(\text{Row 2})$$

$$\mu_3 = \text{mean}(\text{Row 3})$$

$$\sigma = \sqrt{\frac{1}{d} \sum (z - \mu)^2}$$

$$H_{\text{final}} = \begin{bmatrix} -1.220, 0.355, -0.200, 1.065 \\ -1.594, 0.205, 0.519, 0.870 \\ -0.588, -1.448, 0.963, 1.073 \end{bmatrix} \begin{matrix} 0 \\ 1 \\ 2 \end{matrix}$$

Predicting the Next words (Language Model)

$$P(\text{next token} | \text{Input Sequence}) = \text{Softmax}(\dots)$$

$$H_{\text{final}} W_{\text{vocab}} + b_{\text{vocab}}$$

3x4
Sequence
Length

dims

size = 5

= (word [morning,
day, night, evening,
today])

W_{vocab} in this case is

$$\begin{array}{c} 4 \times 5 \\ \swarrow \quad \searrow \\ \text{dims} \quad \text{vocab} = \text{Size} \end{array}$$

$$W_{\text{vocab}} = \begin{matrix} & \text{morning} & \text{day} & \text{night} & \text{evening} & \text{today} \\ \begin{bmatrix} 0.1 & 0.2 & 0 & -0.1 & 0.05 \\ 0 & 0.1 & 0.2 & 0 & 0 \\ 0.5 & 0 & 0.1 & 0.2 & 0 \\ 0 & 0.05 & 0 & 0.1 & 0.1 \end{bmatrix} \end{matrix}$$

$\text{bias} = b_{\text{vocab}} = 0$

Logit for last token "Nice" :

$$h_{2 \rightarrow \text{last}} = [-0.588, -1.448, 0.963, 1.073]$$

$$(1 \times 4)(4 \times 5) = (1 \times 5)$$

$$\text{Logits} = h_2 \times W_{\text{vocab}} =$$

$$\begin{aligned} & -0.588 \times 0.1 + 0 + 0.963 \times 0.5 + 0 = -0.01065 \\ & -0.588 \times 0.2 + (-1.448 \times 0.1) + 0 + 1.073 \times 0.05 \\ & -0.588 \times (-0.1) + (-1.448 \times 0.2) + 0.963 \times 0.1 + 0 \\ & -0.588 \times -0.1 + 0 + 0.963 \times 0.2 + 1.073 \times 0.1 \\ & -0.588 \times 0.05 + 0 + 0 + 1.073 \times 0.1 \end{aligned}$$

$$\text{Logits} = \begin{bmatrix} -0.01065 & -0.26075 & -0.1933 & 0.3587 & 0.073 \end{bmatrix}$$

Morning day night evening today

Display Softmax

$$= [0.192, 0.158, 0.168, 0.278, 0.210]$$

So the highest probability is

0.35587 Evening.